AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW

RESOURCE LOCAL LANGUAGES

R26-DS-11



Meet Our Team



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THE PROBLEM


Imagine rain pounding Sri Lanka for days.
invideo/it22129758
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Problem, Stakeholders, Pain Point

Problem

- During disasters such as floods and fires, emergency reports arrive through calls and SMS.
- Reports come in Sinhala, Tamil, and English.
- Multiple people may report the same incident in completely different words.

Operator Challenge

- Understand multilingual, sometimes panicked speech.
- Judge how urgent the report really is.
- Detect whether it is a duplicate of an already reported incident.

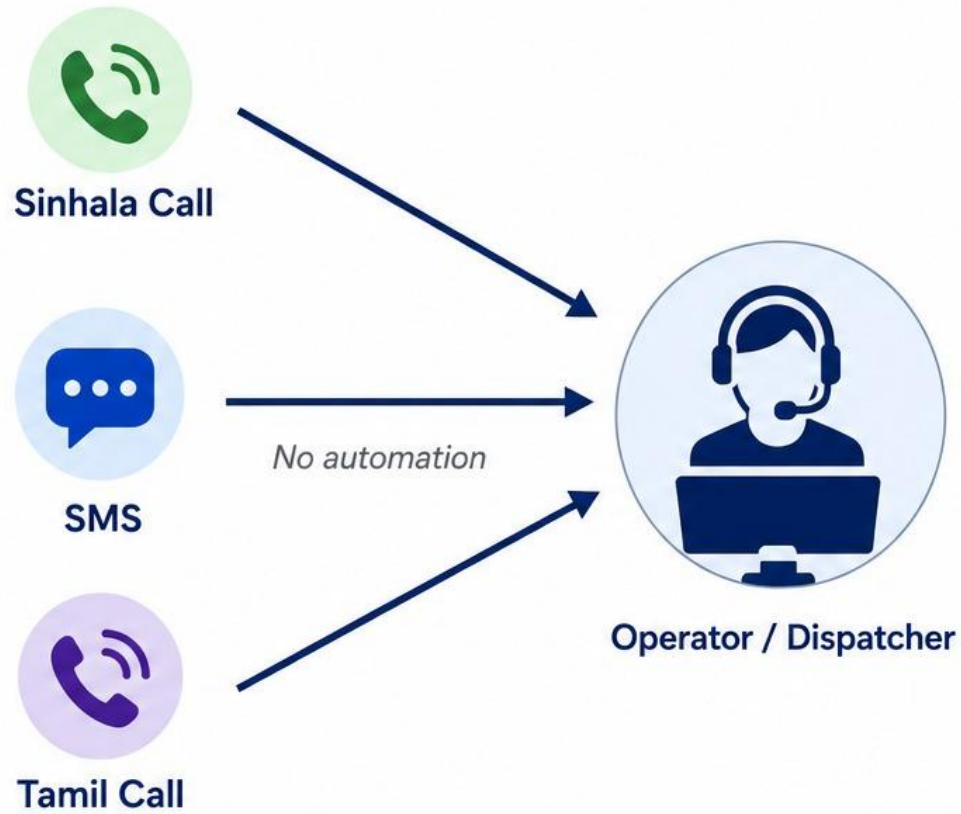
Stakeholders

- Emergency call operators / dispatchers
- Callers and victims
- Disaster Management Centre (DMC)
- First responders

Pain Point

None of those three judgments is automated today.

This can lead to duplicate dispatches for the same incident, or a critical call receiving the same triage attention as a routine one.



one incident, three uncoordinated reports.

Research Gap + Objectives

Research Gap

Existing emergency-AI research handles these pieces in isolation — urgency detection alone, or single-channel intake alone. There is still no integrated approach that combines multilingual voice and SMS intake with cross-source deduplication for low-resource disaster response.

Fragmented Prior Work (Handled in Isolation)

Voice Intake & Urgency
Detect urgency from voice only.

SMS Intake
Process SMS reports only.

Deduplication
Match and deduplicate within one source alone.

Objectives

C1
Classify caller urgency from voice and text, validated honestly using real government disaster calls, not only simulated data.

C2
Understand multilingual speech in real time and ask adaptive follow-up questions until the report is dispatch-ready.

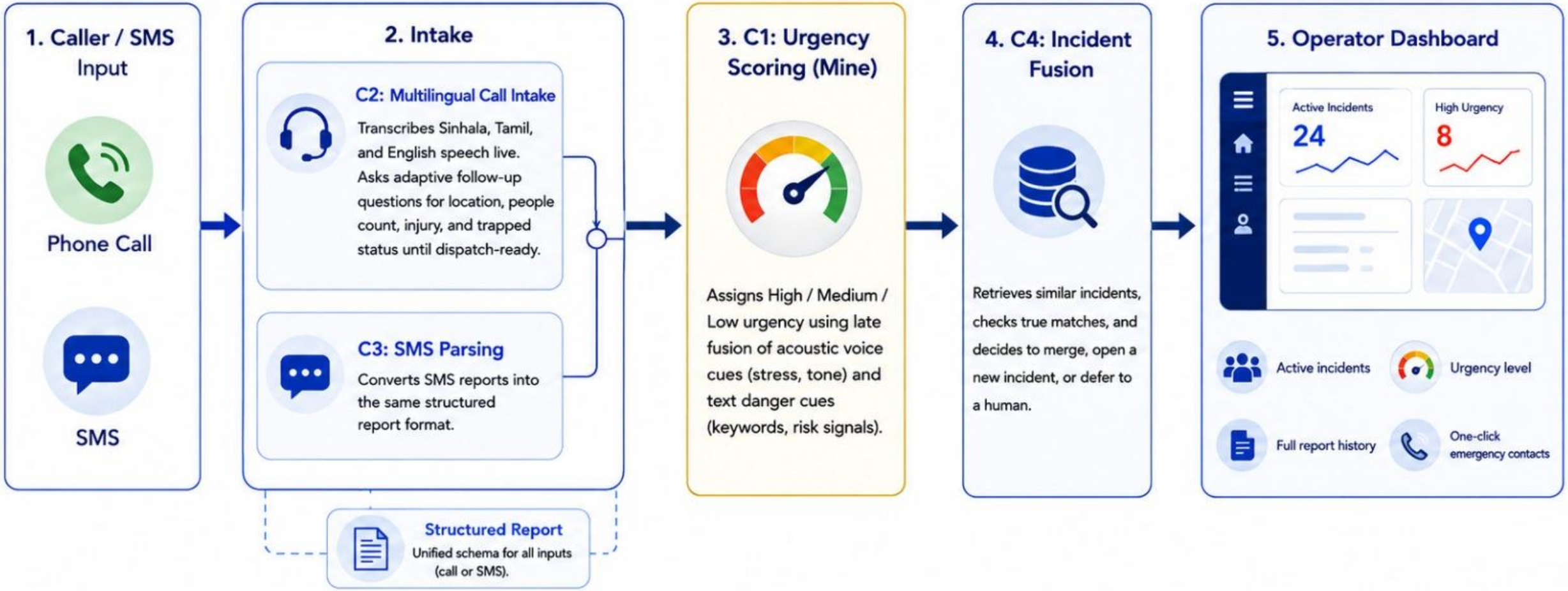
C3
Extract structured, actionable intelligence from SMS emergency reports.

C4
Fuse reports from multiple callers and channels into one verified incident.

Integrated System

Four components working together to deliver one verified, actionable incident for low-resource disaster response.

Solution Overview + Architecture



Unified pipeline from multilingual calls and SMS to verified incident handling.

Integration and Completion Status

Integrated Pipeline

- All four components run as one live, integrated pipeline — not four standalone demos.
- Verified end-to-end this week: a real integration bug was fixed where only a fragment of a multi-turn call narrative reached C4, causing true duplicate reports to be missed.
- Resilience: even a call cut off mid-sentence is still automatically classified and reaches the dashboard within seconds — no report is silently dropped.

Operator Dashboard Improvements

- Shows the full text of every merged report, not just the latest one.
- Flags low-confidence matches for human review.
- Provides one-click dial for the correct emergency contact.

C1 Research Validation

- The urgency model was checked against six independently tested tuning strategies.
- All were validated on simulated data only.
- Any strategy that did not generalize to real calls was rejected.
- The final model was kept honest rather than overfit.

Completion Status

- C1 — Integrated
- C2 — Integrated
- C3 — Integrated
- C4 — Integrated

2 known limitations documented.

Demo-ready end to end.

Two known limitations are documented, but they are not integration-blocking.

Evaluation Evidence and Non-Functional Requirements

Evaluation Evidence

We also evaluated the individual prototypes.

C1 – Urgency Classification



The urgency classification component analyses emergency calls and identifies the urgency level to support faster decision-making.

High | Medium | Low

C2 – Rapid Clarification



Prototype feedback showed that the system can help identify missing critical information quickly and reduce the operator's manual effort when deciding the next question.

C3 – SMS Intelligence



The SMS component extracts structured, actionable intelligence from SMS reports.

C4 – Incident Fusion



The incident fusion component combines information from different sources, identifies related reports, and creates a clear incident record for operators.

Therefore, we are not only showing that the system works; we are also identifying its current limitations and the areas that need further validation.

Non-Functional Requirements

Fast Processing
Real-time processing to support time-critical emergency decisions.

Multilingual Usability
Supports Sinhala, Tamil, and English seamlessly.

Robustness to Noisy Calls
Performs reliably even with background noise and panicked speech.

Low-Bandwidth Communication
Works effectively over unstable networks and low bandwidth.

Explainability
Provides clear reasons and evidence for decisions.

Operator-Friendly Outputs
Clear, concise, and actionable information for operators.

Safety
Designed to minimize harm and prioritize life-critical incidents.

Real-Time Emergency Operations
Built to support continuous, real-time emergency response workflows.

Commercialization + Demo Roadmap

Commercialization Path

Deployable as a triage-and-fusion layer for national/regional emergency call centers, DMC operations, or telecom-partnered SMS emergency gateways.

Generalizable Approach

The approach generalizes to any multilingual, low-resource emergency-response setting, not just Sri Lanka.

Now: Live Demo



1. Multilingual Call

A caller reports an incident in Sinhala / Tamil / English.

2. Clarification (C2)

The system asks follow-up questions and collects all critical information.

3. Urgency Scoring (C1)

The report is scored for urgency: High / Medium / Low.

4. SMS Report

An SMS about the same incident arrives in any language.

5. Auto Fusion (C4)

The SMS report is automatically merged with the call report as the same incident.

6. Operator Console

Both reports appear together with full history and urgency, ready for dispatch.

Transition to Live Demo

Let's see it in action. We will now show a live demo of a multilingual call going through clarification and urgency scoring, an SMS about the same incident merging automatically, and both appearing together on the operator console. **Ready to dispatch.**

- **C1** Integrated
- **C2** Integrated
- **C3** Integrated
- **C4** Integrated

2 known limitations documented.

Addressing PP1 Comments

How each member updated the work after PP1 feedback

IT Number	PP1 feedback response / update
IT22504418	• Replaced static rule-based matching with a model-based incident matching pipeline. • Fine-tuned LaBSE retrieves candidate incidents; joint mDeBERTa matches a new report against full incident history. • A learned policy model decides MERGE / SEPARATE / DEFER. • Location is separated from semantic matching to avoid place-name shortcuts. • Final decisions use report text and accumulated incident evidence.
IT22129758	**Speech-to-Text Evaluation** • Whisper log-probability = confidence, not accuracy; accuracy measured using WER/CER against human transcripts. • Lower WER/CER = better performance. **Dialogue-Aware Clarification Policy Agent** • Adaptive policy decides: ask, confirm, redirect, calm, or end. • Handles uncertainty, silence, stress, and off-topic cases.
IT22263476	• Tested a pure-LLM priority classifier and found it could miss life-threatening messages such as “we can’t breathe.” • Used a deterministic protocol for safety-critical priority and LLMs where they are more suitable. • Strengthened contribution with measured findings: 4.5% cross-model agreement for location extraction, rebuilt hallucination benchmark, and a priority bug found and fixed. • For emotional detection, justified using an LLM because distress depends on context and the question, not just keywords. • Expanded scope to limitations found in testing, including multi-emergency handling and concurrent users.
IT22644480	• Increased accuracy by removing noisy word_count feature. • Changed from early fusion to late fusion and switched to a better-performing Sinhala ASR. • Improved critical condition identification by focusing on high-urgency calls. • Used refined features, late fusion, better ASR, and decision threshold tuning to prioritize detecting high urgency.

All PP1 comments were converted into measurable design changes, model choices, or documented limitations.